CLASS 10 — SCIENCE · FORMULA SHEET

Chapter 10: **The Human Eye and the Colourful World** | Quick revision, one page

Structure and working of the eye

- Cornea (most refraction), aqueous humour, iris (controls pupil size), pupil, lens (adjusts focal length), ciliary muscles, retina (image screen), optic nerve.
- Image on the retina is real and inverted; the brain perceives it erect. Accommodation: ciliary muscles change the lens curvature (thick for near, thin for distant vision).
- Normal eye: near point (least distance of distinct vision) $= 25$ cm; far point $=$ infinity. Persistence of vision $\approx \frac{1}{16}$ s.

Defects and correction

- Myopia: far point nearer than infinity; image in front of retina; corrected by concave lens, $f = -$ (far point).
- Hypermetropia: near point farther than 25 cm; image behind the retina; corrected by convex lens.
- Presbyopia: weakening of ciliary muscles with age; bifocal lenses. Cataract: clouding of the lens, restored by surgery.
- Power of a lens $P = \frac{1}{f}$ (f in metres), unit dioptre (D); concave $-$, convex $+$.

Dispersion and atmospheric refraction

- Prism: white light splits into VIBGYOR; violet deviates most, red least (red has the highest speed in glass). Newton: a second prism recombines the colours.
- Rainbow: dispersion + internal reflection + refraction in water droplets; Sun behind the observer.
- Atmospheric refraction: twinkling of stars (varying density of air, point source), advance sunrise and delayed sunset by about 2 minutes. Planets do not twinkle (extended source).

Scattering

- Tyndall effect: scattering of light by colloidal particles (visible path of a beam in dusty room, forest canopy).
- Blue sky: shorter wavelengths scattered more; red Sun at sunrise and sunset: longer path in the atmosphere; sky dark for astronauts (no atmosphere). Red light: least scattered, used for danger signals.